

1. PROJECT & TEAM INFORMATION

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models
- **Team ID:** SWTID-2026-7486
- **Team Leader:** Diwan Oli M

2. FUNCTIONAL REQUIREMENTS (FR) These represent the critical operational capabilities the ComicCraft software shall perform.

- **FR1 (User Interface):** The application shall provide an HTML-based form to capture story ideas, character details, and artistic preferences.
- **FR2 (Story Engineering):** The application shall securely connect to Google Gemini models to construct a logical story sequence and corresponding character dialogues.
- **FR3 (Illustration Generation):** The application shall interact with Hugging Face's Stable Diffusion API to autonomously design comic panels based on the AI-generated script.
- **FR4 (Preview Rendering):** The system shall utilize a templating engine (Jinja2) to dynamically construct a web-based reading interface containing the generated texts and images.
- **FR5 (PDF Creation):** The backend shall incorporate a module (FPDF) to programmatically align and stitch the textual and visual elements into a downloadable PDF format.
- **FR6 (Error Resilience):** The system shall feature built-in fallback routines (such as alternate image routing or placeholder generation) to prevent total application failure during API timeouts.

3. NON-FUNCTIONAL REQUIREMENTS (NFR) These dictate the qualitative aspects and technical constraints of the system.

- **NFR1 (Execution Speed):** The complete pipeline—from user click to full 5-panel comic presentation—must optimize network calls to complete within approximately 45 to 60 seconds.

- **NFR2 (Accessibility):** The frontend design must be straightforward, requiring zero prior knowledge of prompt engineering from the user.
- **NFR3 (Backend Concurrency):** The system must be built on an asynchronous framework (FastAPI) to handle internal routing efficiently.
- **NFR4 (Data Security):** Sensitive API tokens for Gemini and Hugging Face must be completely isolated from the application's source code utilizing environment (`.env`) variables.